Fix the Structural Bottleneck: Context Compression via Explicit Information Transmission

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Abstract

Long-context LLM agents often struggle with growing token, memory, and latency costs, making efficient context compression essential for practical deployment. Existing LLM-as-a-compressor methods remain noticeably inferior to using the full context. We find that this gap partly stems from their inability to preserve contextual information effectively. In this work, we revisit context compression from a structural perspective and identify two key bottlenecks in standard LLM-based compressors: limited coordination among compression tokens during information aggregation, and layerwise dilution that weakens useful signals from intermediate hidden states. To address these limitations, we propose **ComprExIT** (Context **Compr**ession via **Ex**plicit **I**nformation **T**ransmission), a new context compression framework based on explicit information transmission. ComprExIT adaptively aggregates features into anchors across frozen LLM layers, then allocates information from anchors to compression slots through a globally coordinated transport plan. Experiments on 12 datasets show that ComprExIT consistently outperforms strong soft-compression baselines, improving average F1 by up to 18.5%, while adding only $\sim$1% trainable parameters and achieving over 2$\times$ faster compression than the fastest baselines. Code is available at `https://github.com/Jiangnan0522/ComprExIT`.

1 Introduction

Large Language Models (LLMs) have become unprecedentedly powerful, and these capabilities come with ever-growing context lengths [Team et al., 2026a, Team, 2026, DeepSeek-AI, 2026]. Long contexts rapidly consume tokens, saturate the effective context window, and enlarge the KV cache, incurring substantial latency and memory overhead [Liu et al., 2025, Xiao et al., 2025]. These costs can ultimately limit the practical performance of LLMs on downstream tasks [Li et al., 2025a, Liu et al., 2024].

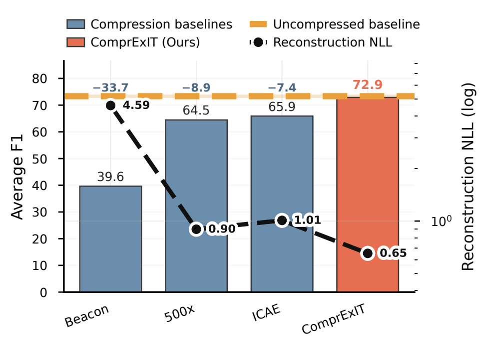

Figure 1: Performance (Average F1) of existing context compression methods and their ability to preserve contextual information (reconstruction negative log-likelihood)

One emerging technique for addressing these problems is *Context Compression* (CC) [Li et al., 2025b], which compresses the original context into a fraction of its length. Current CC methods largely follow the LLM-as-a-compressor paradigm: they repurpose an LLM as a compressor by modifying its internal computation through continual training [Mu et al., 2023, Ge et al., 2024, Li et al., 2025c, Zhang et al., 2025, Deng et al., 2025a]. Specifically, as shown in Figure 2-left, context tokens and a small set of gist tokens are fed into the LLM, which is trained

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to encode contextual information into these gist tokens. Although many design choices have been explored within this paradigm, existing methods still fall well short of the uncompressed baseline (Figure 1 bars). We hypothesize that a key reason is their limited ability to retain contextual information during compression. To investigate this, we use the gist tokens produced by different compressors as inputs and measure the decoder LLM's reconstruction negative log-likelihood (NLL) on the original context [2], which serves as a proxy for how much information the compressed representation preserves. As shown in Figure 1 (black line), methods with worse downstream performance (lower F1) also yield higher reconstruction NLL, suggesting that insufficient information retention is a critical bottleneck of existing approaches.

Since the current LLM-as-a-compressor paradigm is built on the standard LLM architecture, this gap in information retention calls for a structural re-examination. By tracing how information flows in the compressor LLM, we identify two limitations. In the *width dimension*, LLMs rely on self-attention to condense information from all context tokens into the gist tokens. Yet this process suffers from a **Lack of Allocation**: each gist token gathers information independently, without coordinating with the others about which parts of the context they represent. Consequently, multiple gist tokens may focus on the same regions while others remain under-covered, leading to information loss. In the *depth dimension*, LLMs propagate information in a layer-wise accumulative manner. As forward propagation proceeds, fine-grained evidence captured in earlier layers can become increasingly entangled with later, more abstract, generation-oriented representations [Zhang et al., 2024, Skean et al., 2025]. As a result, early-layer features become harder to recover from the final gist-token states, leading to **Information Dilution** [Team et al., 2026b].

The analysis above motivates a different formulation: instead of tuning an LLM into a compressor, we treat it as a *frozen* feature extractor and directly use its hidden states for compression. We propose ComprExIT (Context **Compr**ession via **Ex**plicit **I**nformation **T**ransmission), a new context compression framework that decomposes compression into two stages, each addressing one of the limitations above. The **Widthwise Stage** is built around an optimal-transport-based [Villani, 2008] transmission plan that globally allocates information across tokens and explicitly coordinates how contextual information is distributed to compression slots. The **Depthwise Stage** establishes weighted shortcuts across layers, where fine-grained features extracted at different levels are directly aggregated, thereby mitigating information dilution across layers.

Our experiments across 12 datasets show that ComprExIT significantly outperforms state-of-the-art CC methods, with an average F1 improvement of 18.5%. It is also lightweight and efficient, adding only $\sim$1% parameters while compressing over 2$\times$ faster than the fastest baselines. Further analysis shows that our key design components improve information preservation and thereby boost compression performance.

We summarize our contributions as follows:

- We analyze why existing LLM-as-a-compressor methods lose contextual information, and identify two key bottlenecks: *lack of allocation* in the width dimension and *information dilution* in the depth dimension.
- We propose ComprExIT, a lightweight context compression framework that treats a frozen LLM as a feature extractor and addresses these bottlenecks with a coordinated widthwise transmission plan and depthwise layer aggregation.
- We show that ComprExIT consistently outperforms prior context compression methods on 12 datasets, while remaining parameter-efficient and faster at compression.

2 Related Work

Existing context compression methods can be divided into two categories: token pruning and soft context compression, which operate in discrete and continuous spaces respectively.

Token pruning. Token pruning methods reduce context length by pruning input tokens according to estimated importance. Representative approaches include SelectiveContext [Li et al., 2023] and LLMLingua-style methods [Jiang et al., 2023, 2024], which score tokens/spans using a language model (e.g., via self-information or related salience measures) and retain only the most informative parts of the context. LLMLingua-2 [Pan et al., 2024] instead distills a lightweight classifier from a stronger LLM (e.g., GPT-4) to decide which tokens to keep. EFPC [Cao et al., 2025] further unifies task-aware and task-agnostic compression within a single framework. Despite their efficiency

[2] Experimental details are in Appendix A.1.1.

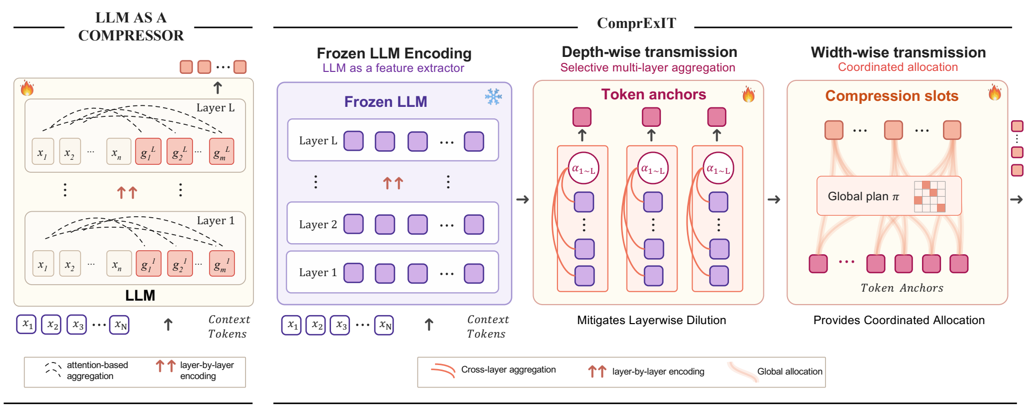


Figure 2: A comparison between existing LLM-as-a-compressor methods (left) and ComprExIT (right). **Existing methods** introduce gist tokens that are iteratively encoded across layers by the self-attention in the LLMs. **ComprExIT** instead leverages the layerwise hidden states of the context tokens encoded in a forward pass. The hidden states are selectively aggregated into token anchors, which are then transmitted to the compression tokens through a coordinated transmission plan.

gains, hard methods remain discrete and lossy, often facing bounded compression ratios and limited expressiveness compared with soft compression in continuous space.

Soft context compression. Soft context compression offers greater flexibility and expressiveness than discrete token removal at high compression ratios, and our method belongs to this category. Wingate et al. presented an early attempt to compress context tokens into a single vector, focusing on representation learning. AutoCompressor [Chevalier et al., 2023] builds LLMs into compressors inspired by the Recurrent Memory Transformer [Bulatov et al., 2022], accumulatively compressing contexts into summary tokens. Mu et al. introduces gist tokens as an information bottleneck by modifying attention masks, establishing the LLM-as-a-compressor paradigm. ICAE [Ge et al., 2024] further simplifies this paradigm into an encoder–decoder framework, which has since become a widely adopted paradigm. Building upon ICAE, 500× [Li et al., 2025c], Activation Beacon [Zhang et al., 2025], and UniGist [Deng et al., 2025a] pass cached key–value states to the decoder, together with carefully designed gist-token attention masks, to obtain more informative compressed representations. Deng et al. systematically study the effects of different design components within the encoder–decoder compression framework. EPL [Zhao et al., 2025] further improves compression performance by adjusting the positional encodings of gist tokens. xRAG [Cheng et al., 2024] explores converting representations from text embedding models into compression tokens. SAC [Liu et al., 2026] examines the necessity of the auto-encoding training objective, demonstrating that effective compression can be achieved using only a text completion objective. Tang et al. [2026] dynamically assign compression budget through coarse-to-fine compression. Different from the existing methods that use the internal self-attention mechanisms of LLMs to perform compression, our method adopts a fundamentally different paradigm by decoupling compression from the LLM architecture and formulating it as an explicit information transmission problem over frozen hidden states.

3 Preliminary

Context compression formulation. Context compression consists of a compressor and a decoder. Given a context sequence $\boldsymbol{x} = (x_1, \ldots, x_N) \in \mathbb{I}^N$, the compressor produces a compact representation $Z = f_\phi(\boldsymbol{x}) \in \mathbb{R}^{K \times d}$ with $K \ll N$. The decoder then conditions on Z to generate outputs.

LLM-as-a-compressor. Most existing methods instantiate f_ϕ by turning an LLM into a compressor. Given a context sequence $\boldsymbol{x} = (x_1, \ldots, x_N)$ and K learnable compression embeddings $G = (\boldsymbol{g}_1, \ldots, \boldsymbol{g}_K) \in \mathbb{R}^{K \times d}$, often called gist tokens, the model feeds the concatenated hidden-state sequence $[\mathrm{Emb}(\boldsymbol{x}); G]$ into a Transformer:

$$H^{(0)} = [\mathrm{Emb}(\boldsymbol{x}); G] \in \mathbb{R}^{(N+K) \times d}, \qquad H^{(\ell)} = \mathrm{Block}_\ell\big(H^{(\ell-1)}\big), \quad \ell = 1, \ldots, L. \tag{1}$$

Here, $H^{(\ell)}$ contains hidden states for all positions in the concatenated sequence. The compressed representation is defined by the compression-token positions in the last-layer output:

$$H^{(L)} = \big[H^{(L)}_{\mathrm{ctx}}; H^{(L)}_G\big], \qquad Z = H^{(L)}_G. \tag{2}$$

Thus, compression is constrained by the same mechanisms used in LLM computation: self-attention for widthwise aggregation and residual layer propagation for depthwise information transfer.

4 Limitations and Observations

Widthwise limitation: lack of allocation. At ℓ-th layer, the attention from gist tokens to context tokens can be written as

$$A^{(\ell)} = \mathrm{softmax}\left(\frac{Q_G^{\ell}(K_X^{\ell})^{\top}}{\sqrt{d_h}}\right) \in \mathbb{R}^{K\times N}. \tag{3}$$

Each row $A_{i,:}^{(\ell)}$ is normalized independently, meaning that each gist token decides where to attend without considering the choices of the others. As a result, standard self-attention provides no mechanism to allocate how gist tokens absorb information across the context: several tokens may concentrate on the same region, while other regions remain weakly covered. To test whether this happens in practice, we analyze the attention distributions of gist tokens during compression [3], which reveal where each token gathers information from. As shown in Figure 3, the attention distributions of different gist tokens are highly correlated, indicating that they often draw information from similar parts of the context. Moreover, the aggregation matrix exhibits a steep singular-value decay, which points to low diversity in the collection behaviour. Together, these findings suggest that many gist tokens capture redundant rather than complementary content. We therefore attribute this pattern to a lack of allocation in self-attention: without explicit coordination, gist tokens cannot spread out to cover distinct context regions and instead collapse onto the same salient areas.

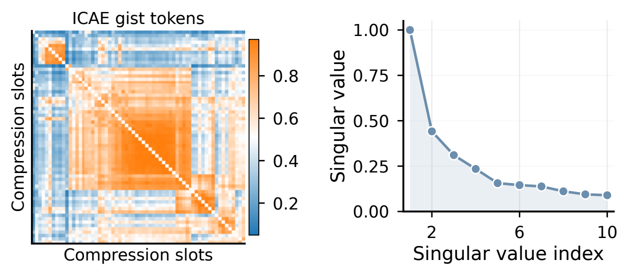


Figure 3: Two views of gist-token aggregation patterns in ICAE: Pearson correlation between gist-token attention distributions and the singular-value spectrum of the attention aggregation matrix.

Depthwise limitation: layerwise dilution. Let $G^{(\ell)} = H_G^{(\ell)}$ denote the compression-token states at layer ℓ. In most LLM-as-a-compressor methods, the decoder only receives the final representation $Z = G^{(L)}$. This design implicitly assumes that task-relevant information gathered at earlier layers is preserved through all subsequent updates. In practice, however, features that are salient at an intermediate layer can be attenuated as the network continues forwarding. To characterize how much usable information remains at depth ℓ, given the task query q and the target output y, we define

$$S_\ell = \max_{\psi\in\Psi} \mathrm{Score}\left(p_\theta(y \mid q, \psi(G^{(\ell)}))\right), \tag{4}$$

where ψ is a lightweight readout or compression module, and $\mathrm{Score}(\cdot)$ denotes the downstream task metric. If S_ℓ reaches its maximum at some intermediate layer and then drops as depth approaches L, this indicates a layerwise dilution bottleneck: information that is accessible earlier is not fully retained in the final state. Motivated by this hypothesis, we examine how information evolves along both the width and depth dimensions when an LLM is tuned as a compressor. We train a probing compressor that uses the hidden states from a single chosen LLM layer. The compressor consists of $4\times$ mean pooling followed by a 2-layer MLP. We train the models on SQuAD and HotpotQA and report the decoder F1. As shown in Figure 4, hidden states from many layers provide useful compression features, but performance is not monotonic across layers, and the final layer performs the worst. This supports the view that useful features formed in earlier layers are gradually diluted rather than faithfully preserved through subsequent transformations. Moreover, contributions of layers are

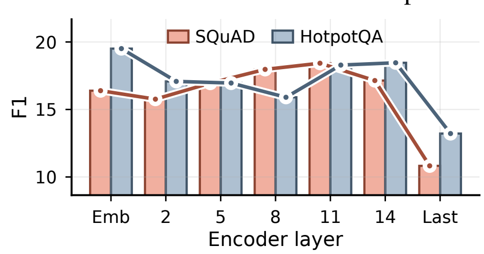


Figure 4: Performance of using each LLM layer for compression.

[3] Experiemnt specifications for examining the allocation patterns can be found in Appendix A.1.3.

task-dependent: early and late layers contribute more to reasoning-intensive QA, whereas middle layers are more effective for extractive QA. These suggest that a strong compression method should both preserve and adaptively exploit features from different layers according to the context and task.

5 Method

We present our method in computational order: from depthwise to widthwise computation.

5.1 Depthwise Transmission to Token Anchors

We first address the depthwise limitation discussed above, namely the layerwise dilution bottleneck. In LLM-as-a-compressor methods, useful information must remain accessible after repeated layer updates before it reaches the final compression states, even though our preliminary analysis suggests that the accessible utility can peak at intermediate layers. Therefore, we directly read from hidden states across depths and construct a token anchor at each token position by adaptively selecting and aggregating multi-layer features. Given the hidden states $\{\boldsymbol{h}_t^{(\ell)}\}_{\ell=1}^{L}$ of token t, we first compute a lightweight structural context by mixing layer representations:

$$\boldsymbol{c}_t = \sum_{\ell=1}^{L} \pi_\ell \boldsymbol{h}_t^{(\ell)}, \qquad \sum_{\ell=1}^{L} \pi_\ell = 1, \tag{5}$$

where π_ℓ denotes learned layer-level structure priors, inspired by Denseformer [Pagliardini et al., 2024]. Conditioned on this context, we perform a token-wise gating attention over layers:

$$\tilde{\boldsymbol{h}}_t = \sum_{\ell=1}^{L} \mathrm{softmax}_\ell \left(\frac{\langle W_q \boldsymbol{c}_t,\ W_k \boldsymbol{h}_t^{(\ell)} + \boldsymbol{e}_\ell \rangle}{\tau} \right) W_v \boldsymbol{h}_t^{(\ell)}, \tag{6}$$

where W_q, W_k, and W_v are learnable query, key, and value projection matrices, $\boldsymbol{e}_\ell$ is a learnable layer embedding that encodes the layer index, and τ is the temperature. The resulting $\tilde{\boldsymbol{h}}_t$ serves as the token anchor. It can be viewed as a gated multi-layer readout from the frozen LLM to the compression interface, rather than an accumulated state that must survive until the last layer. In this way, features that are useful at intermediate depths remain directly accessible to the compressor, which mitigates the layerwise dilution bottleneck. Moreover, because the gating is conditioned on each token, ComprExIT can preserve different granularities of information for different tokens and tasks.

5.2 Widthwise Transmission to Compression Slots

Given the token anchors aggregated across layers, we next address the widthwise limitation discussed above, namely the lack of allocation among compression tokens. Instead of letting each slot independently retrieve information as in self-attention, we explicitly allocate information from N token anchors to K compression slots through a shared transmission plan. Formally, we seek $\Pi \in \mathbb{R}_+^{N \times K}$, where $\Pi_{t,k}$ denotes how much information token anchor t sends to compression slot k.

Utility matrix. To coordinate all possible sender-slot paths globally, we first construct a *utility matrix* $U_{t,k}$ that measures how beneficial it is to transmit information from anchor t to slot k. We treat each token anchor as a *sender*. For each compression slot, we build a corresponding *receiver* and assign it a local field of senders so that the initial slot semantics remain aligned with the original token order. Concretely, we uniformly partition the token-anchor sequence into local fields $\mathcal{F}_k$, average the anchors within each field to obtain the receiver representation, and then compute the utility matrix using the cosine similarity between projected sender and receiver representations:

$$\boldsymbol{r}_k = \frac{1}{|\mathcal{F}_k|} \sum_{t \in \mathcal{F}_k} \tilde{\boldsymbol{h}}_t, \qquad U_{t,k} = \cos\Big(W_u \tilde{\boldsymbol{h}}_t, W_u \boldsymbol{r}_k \Big) . \tag{7}$$

Here W_u is a learnable projection matrix that maps senders and receivers into a shared utility space. Higher utility indicates that sending information along this path is more likely to preserve useful content for the compressed representation.

Information capacity. Different token anchors need not contribute equally to compression. To reflect their varying contextual importance, we assign each sender a learnable information capacity so that less useful anchors can transmit less mass while salient anchors retain more influence. We predict this capacity with a linear layer followed by a softmax over token anchors:

$$\rho_t = \mathrm{softmax}_t\Big(W_\rho \tilde{\boldsymbol{h}}_t\Big), \qquad t = 1, \ldots, N. \tag{8}$$

Here W_ρ is a learnable scoring matrix that produces the sender-capacity logits.

For the receivers, we use a uniform capacity $\rho_k = \frac{1}{K}$. This gives every compression slot the same total budget, while still allowing important anchors to connect to non-local slots to better preserve long-range dependencies.

Transmission plan. Given the utility matrix and the sender/receiver capacities, we derive the transmission plan by solving the following optimization problem:

$$\min_{\Pi \geq 0} \quad \sum_{t=1}^{N}\sum_{k=1}^{K} \Pi_{t,k}\, C_{t,k} \quad \text{s.t.} \qquad \sum_{k=1}^{K} \Pi_{t,k} = \rho_t, \qquad \forall t,\ \sum_{t=1}^{N} \Pi_{t,k} = \rho_k, \qquad \forall k, \tag{9}$$

where the cost is defined as $C_{t,k} = 1 - U_{t,k}$. Unlike self-attention, whose weights are normalized independently for each slot, this objective couples all sender-slot assignments through shared marginal constraints. The resulting solution is exactly an optimal transport plan [Villani, 2008] between senders and receivers, which directly targets the lack-of-allocation issue identified in the preliminary analysis. We solve the problem with entropy regularization, yielding a strictly convex objective that can be optimized efficiently with the Sinkhorn algorithm [Sinkhorn and Knopp, 1967, Cuturi, 2013]. In practice, for stability and efficiency on long sequences, we run Sinkhorn on fixed-size segments of length T. We use a relatively large segment (e.g., $T = 128$) to preserve broad allocation flexibility while avoiding overly aggressive long-range assignments that may disrupt local semantic order. This produces a globally coordinated yet locally aware transmission plan.

Final representations. Given the transmission plan, each compression slot aggregates projected token-anchor features and is then aligned by a lightweight MLP:

$$\boldsymbol{z}_k = \mathrm{MLP}\left(\sum_{t=1}^{N} \Pi_{t,k}\, W_g \tilde{\boldsymbol{h}}_t\right), \qquad k = 1, \ldots, K. \tag{10}$$

Here W_g is a learnable projection matrix used before slot-wise aggregation. The resulting compressed sequence is denoted by $Z = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_K)$.

6 Experiments and Analysis

6.1 Experimental Setup

Implementation details. In the experiment, we use Llama-3.2-1B-Base and Llama-3.2-3B-Base [Grattafiori et al., 2024] as the base model. For the projection to the decoder's input space, we set the projection hidden size to 256. The base models are trained using BF16 precision. We employ the Sinkhorn algorithm with 30 iterations to approximately solve the entropy-regularized optimal transport problem.

Baselines. We compare with several soft state-of-the-art compression baselines (1) **ICAE** [Ge et al., 2024]: a seminal LLM-as-a-compressor method that trains an LLM to encode compressed information into compression tokens. (2) **500×** [Li et al., 2025c]: introduces an extra design that passes the KV states of the compression tokens to the decoder (3) **Activation Beacon** [Zhang et al., 2025]: introduces the interleaving placement of compression tokens. We also include three non-compression baselines: (4) **Zero-shot [w/ context]** prompts the LLM with context, serving as an untrained baseline with zero information loss; (5)**Zero-shot [w/o context]** prompts the LLM without the context, serving as an untrained baseline with complete information loss; and (6) **Prompt tuning** [Lester et al., 2021] also prepends learnable tokens while retaining full access to the context, and therefore serves as a strong uncompressed baseline in our experiments.

Training and evaluation. Except for the base models evaluated under the inference-only setup, all baselines are trained under the same two-phase procedure. We adopt a standard two-phase training procedure under a question-unaware setting: next-token prediction (NTP) on 1B tokens sampled from SlimPajama [Shen et al., 2023], followed by supervised fine-tuning (SFT) on MRQA [Fisch et al., 2019] with six in-domain and six out-of-domain question-answering datasets. We set the compression ratio to $\times 4$. All models use a context length of 512 tokens in both phases. We report EM and F1, and study additional settings in the ablations.

Table 1: Experimental results on six QA benchmark datasets. Best among compression baselines is **bold**. Our method is colored in . The prompt-tuned uncompressed baseline is colored in .

Methods	SQuAD		NewsQA		TriviaQA		SearchQA		HotpotQA		NQ		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B														
Prompt tuning [w/ context]	71.89	81.09	30.72	45.76	61.04	66.75	64.03	70.64	53.60	69.49	53.03	66.59	55.72	66.72
Zero-shot [w/ context]	16.59	38.76	9.97	26.99	35.50	49.31	5.83	12.39	22.51	39.24	18.14	38.90	18.09	34.27
Zero-shot [w/o context]	2.80	8.83	0.97	3.11	15.63	21.60	4.04	6.64	2.76	7.17	3.37	8.43	4.93	9.29
ICAE [Ge et al., 2024]	36.84	50.21	20.20	33.73	57.59	63.54	67.40	74.43	40.67	57.04	42.26	58.00	44.16	56.16
500x [Li et al., 2025c]	4.65	14.02	1.85	5.27	26.09	32.66	17.44	25.59	5.56	13.44	5.58	13.89	10.20	17.48
Beacon [Zhang et al., 2025]	24.88	39.50	9.12	15.74	0.51	2.15	1.90	9.86	31.23	46.35	21.04	37.05	14.78	25.11
ComprExIT	**51.38**	**68.08**	**30.39**	**49.12**	**64.62**	**70.93**	**71.91**	**78.93**	**49.84**	**68.40**	**45.93**	**63.88**	**52.34**	**66.55**
Llama-3.2-3B														
Prompt tuning [w/ context]	80.39	88.41	35.23	50.98	71.23	76.30	72.04	78.42	55.31	72.65	59.46	73.32	62.28	73.35
Zero-shot [w/ context]	35.85	55.05	16.33	34.89	57.52	67.83	29.40	37.26	38.94	57.24	32.74	52.09	35.13	50.73
Zero-shot [w/o context]	9.14	17.21	2.40	5.78	46.20	52.06	25.43	30.35	8.63	16.03	10.31	18.47	17.02	23.32
ICAE [Ge et al., 2024]	48.51	62.53	27.02	43.04	68.62	74.13	76.43	82.89	47.92	66.16	50.87	66.75	53.23	65.92
Beacon [Zhang et al., 2025]	59.81	72.65	15.46	24.39	1.36	4.72	5.24	9.13	51.55	67.41	45.05	59.56	29.75	39.64
500x [Li et al., 2025c]	52.08	65.56	25.45	40.99	65.77	71.59	69.26	76.28	47.28	65.71	51.32	66.60	51.86	64.46
ComprExIT	**59.26**	**75.68**	**35.73**	**55.23**	**72.97**	**78.86**	**78.00**	**84.40**	**55.92**	**74.15**	**52.23**	**68.95**	**59.02**	**72.88**

Table 2: Key design ablations for ComprExIT on Llama-3.2-1B (16 layers), using the main experiment's setup.

Ablation	Avg. EM	Avg. F1
ComprExIT	52.34	66.55
w/o alloc.	47.69 ↓4.65	61.94 ↓4.61
only emb. layer	28.65 ↓23.69	44.49 ↓22.06
only layer 5	45.13 ↓7.21	61.39 ↓5.16
only layer 9	49.34 ↓3.00	63.89 ↓2.66
only layer 14	44.18 ↓8.16	60.74 ↓5.81
only last layer	36.12 ↓16.22	49.35 ↓17.20

Table 3: Latency analysis results.

Method	Compress (s)	Decode (s)	E2E (s)
Context Length = 512			
Full Context	–	25.04	25.04
ICAE	0.43	10.16	10.59
ComprExIT	**0.18**	**9.83**	**10.01**
Context Length = 2048			
Full Context	–	101.09	101.09
ICAE	1.93	27.37	29.30
ComprExIT	**0.84**	**27.18**	**28.02**

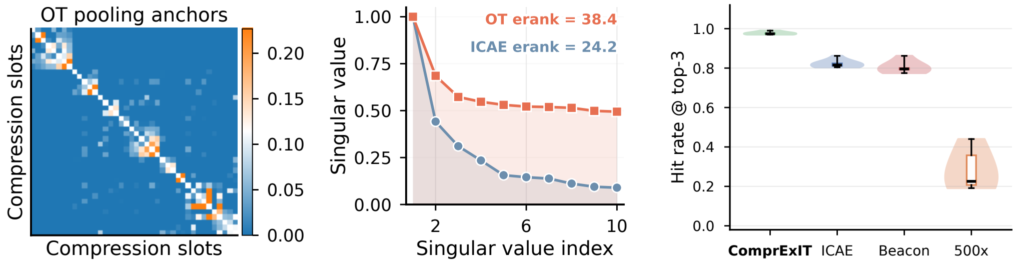
6.2 Main Results

Overall performance. Under the question-unaware training setup described above, Table 1 shows that ComprExIT consistently outperforms all compression baselines on the six in-domain QA benchmarks for both Llama-3.2-1B and Llama-3.2-3B. In average F1, it improves over the strongest compression baseline by 18.50% and 10.56%, respectively, while remaining close to the uncompressed prompt-tuning baseline (within 0.25% on 1B and 0.64% on 3B). Overall, these results show that ComprExIT performs better under the same compression budget.

Out-of-domain performance. Using the same training recipe and experimental setup, we further evaluate on six out-of-domain MRQA datasets in Table 8. ComprExIT again performs best overall, improving average F1 over the strongest compression baseline by 30.97% on Llama-3.2-1B and 14.27% on Llama-3.2-3B, with only a small gap to ICAE on RelationExtraction. Since RelationExtraction has very short contexts (30 tokens on average) and ICAE and 500x employ a fixed budget of 128 compression tokens, they are constrained by a much lower compression ratio on this dataset, thereby yielding better performance. We also evaluate on summarization, fact-checking, and in-context learning in Appendix A.4. ComprExIT outperforms the uncompressed baseline on XSum summarization and FEVER fact-checking while remaining competitive on SST-2 in-context learning. Together, these results suggest that ComprExIT generalizes well to various domains.

6.3 Ablation Studies

We examine our design choices under the same training and evaluation setup as before with Llama-3.2-1B. To isolate the effect of coordinated allocation, we replace the OT-based allocation module with a window attention mechanism that preserves locality but removes global coordination. As shown in Table 2, performance drops consistently across all datasets, indicating that global allocation is critical for effective compression. Next, we remove layer-wise aggregation and use features from only a single LLM layer for compression. Performance degrades under all single-layer settings, with the middle layer (Layer 9) incurring the smallest drop. This result further validates that the design in ComprExIT that selectively aggregates features from all layers benefit compression.



(a) Pearson correlation. (b) Singular-value spectrum. (c) Entity coverage.

Figure 5: Information allocation analysis. **(a)** Pearson correlation and **(b)** singular-value spectrum of allocation patterns. **(c)** Question-relevant entity recall on HotpotQA by TF-IDF rank. Top-10 entities are selected per context, and violin plot shows retrieval consistency across entities.

6.4 Latency Analysis

Table 3 reports the latency breakdown under two context lengths [4]. Compared with ICAE, ComprExIT is more than 2× faster overall in the compression stage across settings. In both settings, the compressed methods are also substantially faster end-to-end than decoding the full context, showing the efficiency gain brought by context compression. These results suggest that ComprExIT forms compressed representations more efficiently, yielding a stronger efficiency-quality trade-off.

6.5 Scalability Analysis

We examine the scalability of our method from three dimensions: higher compression ratios, larger models and longer context.

Higher compression ratios. Under the same setup, we train the Llama-3.2-1B with compression ratios of ×8 and ×16 and report the results in Table 4. ComprExIT outperforms the baseline at both 8× (+17.2% Avg. F1) and 16× (+7.6%). Notably, ComprExIT at 8× compression (58.66) surpasses ICAE at 4× compression (56.16; Table 1). These results show that ComprExIT exhibits consistently strong performance under higher compression ratios.

Larger model sizes. To examine the scalability of ComprExIT on larger models, we train the methods with Llama-3.1-8B under the same setup and report the performance in Table 4. ComprExIT continues to outperform ICAE by a clear margin and even slightly surpasses the uncompressed prompt-tuning baseline in F1 (76.48 vs. 75.92), showing that its advantage persists at larger model scale.

Longer context. We train the Llama-3.2-1B model on three long-context QA datasets, TriviaQA [Joshi et al., 2017], QuALITY [Pang et al., 2022], and NaturalQuestions [Kwiatkowski et al., 2019], with contexts truncated to a maximum length of 8k. The results are shown in Table 4. On this setting, ComprExIT improves over the uncompressed prompt-tuning baseline by 18.0% in F1 and outperforms ICAE by a large margin. This suggests that ComprExIT can preserve task-relevant content under more challenging long-context compression settings.

Table 4: Scalability analysis experiments for ComprExIT on higher compression ratios, larger models, and longer context.

Analysis	**Setting**	**Avg. EM**	**Avg. F1**
Compression ratio	×8 - ICAE	38.72	50.07
	×8 - ComprExIT	**44.83**	**58.66**
	×16 - ICAE	37.23	48.82
	×16 - ComprExIT	**40.01**	**52.53**
Model size	8B - Prompt tuning	**65.03**	75.92
	8B - ICAE	59.24	70.41
	8B - ComprExIT	63.25	**76.48**
Context length	8k - Prompt tuning	27.43	37.27
	8k - ICAE	21.99	30.00
	8k - ComprExIT	**32.46**	**43.97**

6.6 Further Analysis

6.6.1 Information Allocation

Allocation patterns. To further understand the impact cast by coordinated allocation, we analyze the aggregation behaviors of compression tokens [5]. In Figure 5a, compared with the aggregation pattern of ICAE in Figure 3, the consistently lower inter-slot correlations of ComprExIT indicate that different compression tokens absorb complementary information rather than collapsing into

[4] Check Appendix A.1.2 for details of the latency analysis experiment.
[5] Experiemnt specifications for examining the allocation patterns can be found in Appendix A.1.3.

redundancy. In Figure 5b, this same behavior is reflected in the singular-value spectrum: compared with ICAE, ComprExIT exhibits a slower decay and a higher effective rank, showing that its allocation patterns span a richer and less redundant subspace. Taken together, the two views suggest that without global allocation, compression tokens tend to become correlated and low-rank, whereas ComprExIT encourages a more diverse and information-efficient distribution of content across slots.

Coverage of entities. We further examine whether coordinated allocation helps preserve important entities during compression. We extract named entities from each context and rank them by TF-IDF score. For each gist token, we identify the top-3 entities it attends to most and take their union within the context. We then compute the coverage of the ten most question-relevant entities by the gist tokens during compression, as shown in Figure 5c. Note that TF-IDF is used here only as a selection heuristic, and its notion of relevance is limited, since entities that are less relevant to one question may still be important for others. ComprExIT achieves the highest average coverage rate, nearly 1.0. More importantly, its thin violin shape indicates that it consistently covers relevant entities across examples, whereas the larger variance of the baselines suggests that they allocate attention more unevenly and sometimes miss important entities in certain contexts. These results indicate that a well-coordinated allocation is better able to preserve critical contextual information during compression.

6.6.2 Layer Selection

General layer preference. Figure 6-left shows that the gating mass concentrates on early and middle layers, with a clear emphasis before layer 10 and consistently low weights on later layers. This suggests that effective compression mainly relies on representations that still preserve lexical and contextual information, while deeper layers are more specialized for generation and therefore less suitable for forming compact, decoder-friendly representations.

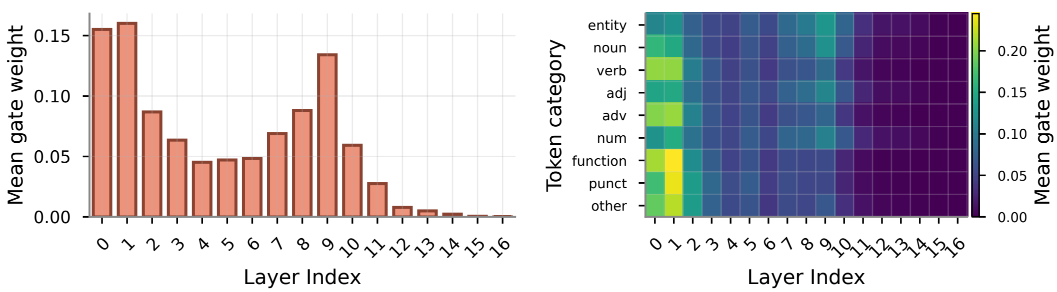


Figure 6: Layer-selection analysis in ComprExIT. *Left*: depth-wise gating weights across layers. *Right*: position–layer heatmap showing token-wise layer preferences.

Token-wise layer preference. The right panel shows a clear dependence on token type. Entity tokens, common nouns, adjectives, and numbers tend to receive higher weights from middle layers, where contextual, relational, and attribute-level information is more salient. In contrast, other tokens are more often routed to the initial layers, indicating that shallow lexical features are usually sufficient for them. Overall, these patterns suggest that ComprExIT adapts layer selection to token type, reserving richer mid-layer features for informative content words and using earlier-layer features for less informative tokens.

7 Conclusion

This work presents ComprExIT, a new paradigm for soft context compression that formulates compression as explicit information transmission over frozen LLM hidden states, mitigating the limitations of layerwise dilution and lack of allocation. Experiments show that it consistently outperforms prior compression methods. More broadly, the proposed paradigm provides a more flexible design space beyond standard attention-based formulations. Therefore, we hope that this work can inspire future research to explore more design choices in this new paradigm and further advance context compression research.

Limitations. First, due to limited computational resources, we do not test ComprExIT on larger backbones (e.g., 32B or 70B models) or substantially longer contexts (e.g., 32k or 128k tokens). Second, our experiments focus on text-only language models. Extending the framework to vision-language models, which also accepts tokenized inputs, is left to future work.

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A Appendix

A.1 Experiment Specification

We present more details of the experiments conducted in our paper.

A.1.1 Measuring Information Loss during Compression

For each method and each compressor, we extract the gist hidden states at layer ℓ, apply the method's own final projection where one exists, and feed them to the frozen decoder as the prefix of the sequence $[\text{BOS}] \,\|\, \mathbf{g}^{(\ell)} \,\|\, \mathrm{Embed}(\mathbf{p}) \,\|\, \mathrm{Embed}(\mathbf{x})$, where $\mathbf{p}$ is the fixed prompt `"Please reproduce the previous passage exactly."` and $\mathbf{x}$ is the original context. We cacluate the standard causal-LM cross-entropy loss on the context tokens only. We then aggregate token-weighted across the dataset, $\widehat{\mathcal{L}}_\ell = \sum_i \mathrm{loss}_i \,/\, \sum_i n_i$, where n_i is the number of non-padded target tokens in example i. Contexts are passages drawn from SQuAD [Rajpurkar et al., 2016] and HotpotQA [Yang et al., 2018] datasets. We use the compression models trained with Llama-3.2-1B in this experiments.

A.1.2 Latency Analysis

We benchmark each method on synthetic batches with fixed context lengths of 512 and 2048, a batch size of 8, a generation length of 128 tokens, and a compression ratio of 4. We use NVIDIA L40S ADA GPU, with 48GB memory. For each sequence, we run 50 untimed warm-up iterations followed by 200 timed iterations, and report the mean and standard deviation. We separately time the *compression* stage (the encoder forward pass that produces the gist) and the *decoding* stage (autoregressive generation of 128 tokens), and report both along with their sum (E2E).

A.1.3 Gist Token Allocation Patterns

We probe whether the K compressed slots of each method actually carry distinct content, or whether they collapse onto similar input regions. We run our experiments on the SQuAD [Rajpurkar et al., 2016] dataset with 1000 random samples. For ICAE we take the compressor's last layer attention map of the gist tokens and average over the attention heads. For ComprExIT, we use the captured transport plan as the allocation scores. In both cases this yields a $[K, N]$ matrix A whose rows are the per-slot distributions over input tokens. The *similarity heatmap* is the $K \times K$ Pearson-correlation matrix between these rows, $\rho(a_i, a_j) = (a_i - \bar{a}_i)^\top (a_j - \bar{a}_j)/(\|a_i - \bar{a}_i\|_2 \|a_j - \bar{a}_j\|_2)$, reordered hierarchically for readability. The *attention spectrum* is the sequence of singular values of A, plus the effective rank $\mathrm{erank}(A) = \exp\big(-\sum_i \tilde{\sigma}_i \log \tilde{\sigma}_i\big)$ with $\tilde{\sigma}_i = \sigma_i / \sum_j \sigma_j$. A high effective rank and a near-diagonal similarity heatmap indicate that the K compressed slots cover the input non-redundantly.

A.2 Datasets

Table 5 shows the statistics of the datasets we use in the experiments.

A.3 Model Inference Implementation

We carefully deal with the position of the padding tokens, by which we find bring improvements to all methods. We keep the no-padding setup during NTP to fully make use of the utility of the GPUs. In SFT, as we stick to the question independent setup, we have to encode the contexts first and feed the context compression tokens along with the question tokens to the decoder. For efficient batch compression, we first apply left-padding to the context tokens first for the compressors. We then concatenate the compression tokens with padded question and answer tokens, and perform an operation to move all the padding tokens to the left most side according to the attention mask. We also create the shifted new attention mask and labels accordingly. This ensure that the generation start from a non-padding token and there is no position gap between the compression tokens and the questions.

A.4 Evaluation on More Tasks.

We further evaluate ComprExIT on summarization, fact-checking, and in-context learning. For summarization, we fine-tune all methods on XSum [Narayan et al., 2018] with Llama-3.2-1B

Table 5: Statistics of the training and evaluation datasets used, including in-domain and out-of-domain datasets. #Train represents the number of training samples and #Validation represents the number of validation samples.

Dataset	Average Question Length	Average Context Length	#Train	#Validation
SQuAD	11	137	86,588	10,507
NewsQA	8	599	74,160	4,212
TriviaQA	16	784	61,688	7,785
SearchQA	17	749	117,384	16,980
HotpotQA	22	232	72,928	5,904
Natural Questions	9	153	104,071	12,836
BioASQ	11	248	–	1,518
DROP	11	243	–	1,501
DuoRC	9	681	–	1,503
RACE	12	349	–	1,502
RelationExtraction	9	30	–	1,500
TextbookQA	11	657	–	1,508

Table 6: ROUGE-1 and ROUGE-L scores on the summarization (XSum) dataset.

Model	ROUGE-1	ROUGE-L
ICAE	0.324	0.257
Prompt Tuning	0.320	0.253
ComprExIT	**0.333**	**0.262**

Table 7: F1 scores on fact checking (FEVER) and in-context learning (SST-2) datasets.

Model	FEVER	SST-2 ICL
ICAE	59.24	66.01
Prompt Tuning	61.97	**72.34**
ComprExIT	**64.39**	69.17

and report the results in Table 6. ComprExIT achieves the best ROUGE-1/ROUGE-L scores, 0.333/0.262, outperforming both ICAE and the uncompressed prompt-tuning baseline. This suggests that ComprExIT preserves not only answer-relevant facts but also the broader discourse information needed for abstractive generation. For fact-checking and in-context learning, we directly evaluate the NTP-trained Llama-3.2-3B models on FEVER [Thorne et al., 2018] and SST-2 [Socher et al., 2013], respectively, as shown in Table 7. On FEVER, ComprExIT achieves the best score, 64.39, surpassing the prompt-tuning baseline by 3.91% and ICAE by 8.69%. On SST-2 ICL, ComprExIT obtains 69.17, improving over ICAE by 4.79% and remaining competitive with the prompt-tuning baseline (72.34). Overall, these results show that ComprExIT generalizes consistently well beyond QA.

A.5 Out-of-Domain Results

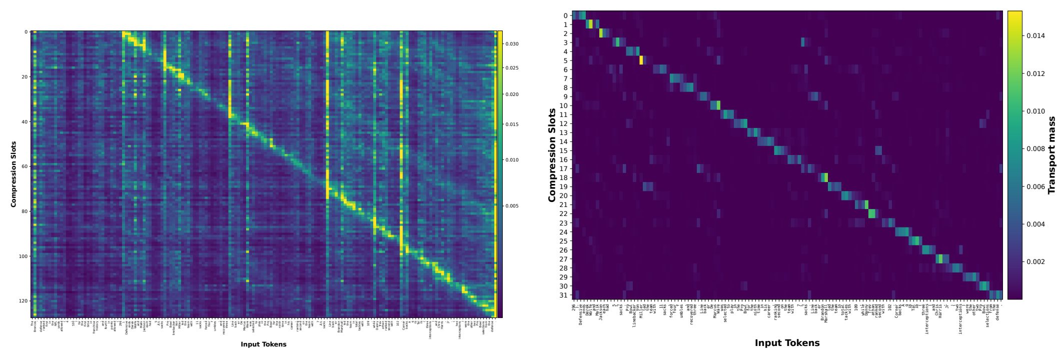
Table 8 reports the full results on the six out-of-domain MRQA benchmarks. Overall, ComprExIT remains the strongest compression method across both Llama-3.2-1B-Base and Llama-3.2-3B-Base, showing that its learned compressed representations transfer well beyond the training domains. We note that ICAE is competitive on RelationExtraction, whose contexts are very short on average. Because prior methods use a fixed budget of 128 compression tokens, this dataset effectively gives them a much smaller compression ratio, which partially explains the smaller gap on that benchmark.

A.6 The Allocation Matrix

Figure 7b visualizes the learned width-wise transmission plan in ComprExIT. Compared with existing methods (e.g., ICAE in Figure 7a), ComprExIT exhibits a more structured allocation pattern: each compression slot is softly anchored to a contiguous local region of input tokens, while still retaining non-negligible connections to distant tokens. This enables ComprExIT to preserve the semantic order of the context without sacrificing the ability to capture long-range dependencies. In contrast, although ICAE tries to maintain an order bias (there is a diagonal pattern of high attention values in Figure Figure 7a), its compression-token attention patterns are weakly localized. Moreover, it can be observed that some important tokens (e.g. entity “Marcus", number “4.") are ignored collectively by most of the compression tokens, but some entities (e.g. Broncos) are overly focused. We attribute these observations to the lack of coordinated allocation.

Table 8: Experimental results on six **out-of-domain** MRQA benchmark datasets. Out-of-domain: these datasets are not included in the training data. Best among compression baselines is **bold**. Our method is colored in . The prompt-tuned uncompressed baseline is colored in . Note that ICAE and 500x have smaller compression ratio on samples of context length shorter than 512 because they are trained with a fixed set (128) of compression tokens. Particularly, on RelationExtraction dataset whose average context length is only 30, they share overly sufficient compression bandwith.

Methods	RelationExtraction		BioASQ		TextbookQA		DuoRC		DROP		RACE		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B-Base														
Prompt tuning [w/ context]	69.91	81.15	65.96	78.80	52.10	59.61	38.51	47.40	32.54	43.14	33.68	46.30	48.78	59.40
Zero-shot [w/ context]	27.48	47.87	6.98	28.84	18.16	29.97	17.46	30.54	18.43	32.30	10.53	22.71	16.51	32.71
Zero-shot [w/o context]	5.09	10.33	14.76	20.56	11.11	16.03	0.60	2.55	14.84	19.73	0.45	2.88	7.81	12.01
ICAE [Ge et al., 2024]	**62.28**	**75.97**	42.62	54.09	28.21	34.60	12.59	18.80	24.15	33.23	9.50	16.95	29.89	38.94
500x [Li et al., 2025c]	6.31	16.30	22.54	35.52	16.77	24.40	1.07	3.91	8.18	18.32	1.93	5.70	9.47	17.36
Beacon [Zhang et al., 2025]	33.18	48.82	32.25	47.84	5.92	10.34	2.40	4.90	21.82	29.24	15.88	25.35	18.57	27.75
ComprExIT	59.46	74.96	**48.40**	**63.74**	**41.65**	**50.73**	**28.78**	**38.77**	**29.01**	**39.86**	**23.00**	**37.95**	**38.38**	**51.00**
Llama-3.2-3B-Base														
Prompt tuning [w/ context]	77.37	86.70	67.62	81.50	60.48	68.75	45.50	55.63	48.84	57.18	43.62	56.35	57.24	67.68
Zero-shot [w/ context]	54.24	68.46	43.22	64.31	36.46	46.66	28.31	42.36	23.69	37.45	22.70	38.16	34.77	49.57
Zero-shot [w/o context]	14.28	21.03	20.94	25.76	23.29	28.89	2.07	4.44	16.17	21.26	1.78	5.65	13.09	17.84
ICAE [Ge et al., 2024]	**70.66**	**82.10**	55.85	67.21	42.25	50.70	23.65	32.81	35.33	44.89	21.51	33.85	41.54	51.93
500x [Li et al., 2025c]	65.13	79.67	47.67	60.82	29.41	38.10	22.25	30.91	29.74	39.12	19.14	30.58	35.56	46.53
Beacon [Zhang et al., 2025]	58.92	71.62	54.52	66.86	10.98	15.68	4.73	7.95	39.39	50.55	30.71	43.95	33.21	42.77
ComprExIT	65.31	77.25	**59.64**	**73.87**	**56.09**	**64.50**	**34.31**	**44.98**	**43.58**	**53.16**	**31.60**	**47.69**	**47.19**	**59.34**



(a) ICAE

(b) ComprExIT

Figure 7: Last-layer attention heatmap of gist tokens produced by ICAE (left) and width-wise information transmission plan in ComprExIT (right).

A.7 Distribution Mismatch Induced by Layerwise Dillution

In this subsection, we formalize the potential mismatch introduced by layerwise dilution. The key idea is that if the compression-token representations drift slightly at each layer, then these small discrepancies can accumulate with depth and enlarge the mismatch between the final compressor output and the representation space expected by the decoder.

Let $p(Z^{(\ell)})$ denote the empirical distribution of compression-token representations $\{\boldsymbol{z}_{\boldsymbol{k}}^{(\boldsymbol{\ell})}\}_{k=1}^{K}$ at layer ℓ, and let p_{dec} denote the representation distribution expected by the decoder. We measure this mismatch using the Wasserstein distance $\mathcal{W}$.

Proposition A.1 (Accumulation of layerwise drift)**.** *Assume $p(Z^{(0)}), \ldots, p(Z^{(L)})$ and p_{dec} all have finite first moments. Define the final compressor–decoder mismatch as*

$$\mathcal{M} := \mathcal{W}\Big(p(Z^{(L)}),\, p_{dec}\Big)\,. \tag{11}$$

Then

$$\mathcal{M} \le \sum_{\ell=0}^{L-1} \mathcal{W}\Big(p(Z^{(\ell+1)}),\, p(Z^{(\ell)})\Big) + \mathcal{W}\Big(p(Z^{(0)}),\, p_{dec}\Big)\,. \tag{12}$$

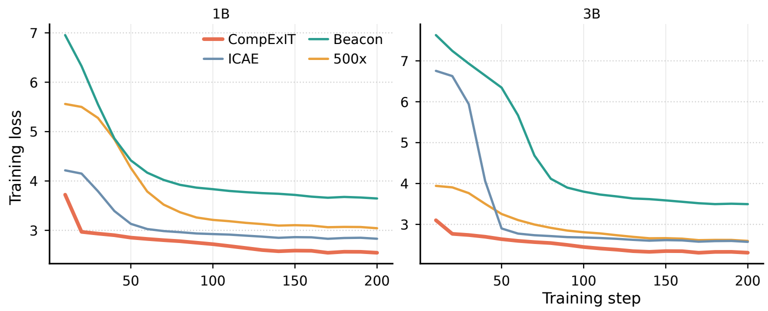


Figure 8: The training curves of baseline methods and ComprExIT under the next-token prediction task.

Proof. By the triangle inequality for the Wasserstein distance [Clement and Desch, 2008],

$$\mathcal{W}\Big(p(Z^{(L)}),\, p_{\text{dec}}\Big) \leq \mathcal{W}\Big(p(Z^{(L)}),\, p(Z^{(L-1)})\Big) + \mathcal{W}\Big(p(Z^{(L-1)}),\, p_{\text{dec}}\Big)\,. \tag{13}$$

Applying the same inequality recursively to the second term yields

$$\mathcal{W}\Big(p(Z^{(L-1)}),\, p_{\text{dec}}\Big) \leq \mathcal{W}\Big(p(Z^{(L-1)}),\, p(Z^{(L-2)})\Big) + \mathcal{W}\Big(p(Z^{(L-2)}),\, p_{\text{dec}}\Big)\,, \tag{14}$$

and continuing this expansion down to layer 0 gives

$$\mathcal{M} \leq \sum_{\ell=0}^{L-1} \mathcal{W}\Big(p(Z^{(\ell+1)}),\, p(Z^{(\ell)})\Big) + \mathcal{W}\Big(p(Z^{(0)}),\, p_{\text{dec}}\Big)\,. \tag{15}$$

This proves the claim. □

The proposition shows that the final mismatch is upper-bounded by two terms: the initial mismatch at the compressor input, and the cumulative layerwise drift accrued across the encoder. Therefore, even if each layer introduces only a modest shift, the total drift can still grow with depth, increase $\mathcal{M}$, and make the compression objective harder to optimize. Together with the extensive experiments in Sections 3 and Appendix A, this provides both theoretical and empirical support for the role of information dilution.

A.8 Training

A.8.1 Optimization Behavior

We plot the NTP training curves in Figure 8. ComprExIT starts from the lowest loss and converges faster to the best plateau, indicating a smaller compressor-decoder mismatch and easier optimization. This supports our core claim that layer aggregation and coordinated allocation produce compressed states that preserve useful information while remaining decoder-friendly.

A.8.2 Training Details

Tables 9 to 11 present the hyperparameters we use when training ComprExIT and the baseline methods.

A.9 Broader Impact

This work studies learned context compression for long-context language models. Its primary potential benefit is improved efficiency: effective compression can reduce memory usage, latency, and computational cost while preserving salient information from the original context. In turn, this may make document-grounded models more accessible to researchers and practitioners with limited hardware resources, and may reduce the energy required to deploy such systems at scale. More efficient long-context modeling may also broaden the practical use of language models in settings where rapid processing of long documents is important, such as literature review, writing assistance, and educational tools. Gist tokens can also function as encrypted memory or a communication medium for LLM agents while remaining unreadable to humans.

Table 9: NTP Training configuration for ComprExIT.

Item	ComprExIT
Run & setup	
Dataset	SlimPajama-6B
Number of Tokens	1 Billion
Device	4× NVIDIA-A100
Precision	Bfloat16
Sequence & compression	
Context length	512
Generation length	128
Compression ratio	4
Optimal-Transport (OT)	
OT window size	128
OT iterations	30
OT projection dimension	256
Layerwise gate hidden size	256
Optimization	
Learning rate	1×10^{-4}
Warmup ratio	0.05
Max grad norm	20.0
Batch size (per device)	16
Gradient accumulation steps	32
Actual Batch Size	2048
Epochs	1

Table 10: NTP Training configuration for ICAE and 500x.

Item	ICAE	500x
Run & setup		
Dataset	SlimPajama-6B	SlimPajama-6B
Number of Tokens	1 Billion	1 Billion
Device	4× NVIDIA-A100	4× NVIDIA-A100
Precision	Bfloat16	Bfloat16
Sequence & compression		
Context length	512	512
Generation length	128	128
Compression ratio	4	4
Memory		
Number of memory tokens	128	128
Optimization		
Learning rate	3×10^{-5}	3×10^{-5}
Warmup ratio	0.05	0.05
Max grad norm	20.0	20.0
Batch size (per device)	16	32
Gradient accumulation steps	16	32
Actual Batch Size	2048	2048
Epochs	1	1
LoRA (compressor)		
Enabled	True	True
Rank r	128	128
Alpha α	32	32
Dropout	0.05	0.05
Bias	none	none
Task type	CAUSAL_LM	CAUSAL_LM

Table 11: NTP Training configuration for Activation Beacon.

Item	Activation Beacon
Run & setup	
Dataset	SlimPajama-6B
Number of Tokens	1 Billion
Device	4× NVIDIA-A100
Precision	Bfloat16
Sequence	
Context length	512
Generation length	128
Activation Beacon configuration	
Beacon enabled	True
Beacon window / stride	64 / 64
Beacon ratio	4
Beacon attention	full-coverage
Attend previous beacons	True
Trainable beacon params	q, k, v
Beacon position	interleave
Grouping by stride	strict
Optimization	
Learning rate	1×10^{-4}
Max grad norm	20.0
Batch size (per device)	16
Gradient accumulation steps	32
Actual Batch Size	2048
Epochs	1

At the same time, context compression introduces important risks. Any compression method may discard rare but critical details, qualifiers, or provenance signals, which can lead to overconfident but incomplete answers. Such failures may be especially concerning in high-stakes applications such as medicine, law, science, or public policy, where omitted evidence can materially affect the correct conclusion. In addition, compressed-context models can still inherit social biases, factual errors, and privacy risks from their pretrained backbone and training data, and efficient processing of long documents could be misused for large-scale analysis of sensitive text. We therefore recommend task-specific evaluation, preserving access to the original context whenever possible, and avoiding deployment as a substitute for expert judgment in safety-critical settings.

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